# OASIS Agentic Pipeline

## Effectiveness & Data Analysis Report

Automated evaluation report generated 2026-06-17.

- MRI scans evaluated: 280 (stratified across 4 classes)
- Clinical cohort: 100 subjects | Longitudinal records: 373
- Acceleration: ONNX Runtime QNN (Snapdragon NPU) → DirectML → CPU fallback
- Language reasoning: local Ollama (no paid API tokens)

Hybrid edge-cloud, Snapdragon-NPU-optimized, zero paid-API multi-agent system for Alzheimer's disease screening on the OASIS datasets.

# 1. Executive Summary

| 98.6% | 98.6% | 100.0% | 100.0% |
|---|---|---|---|
| Overall accuracy | Balanced acc. | Screening sensitivity | Guardrail acc. |

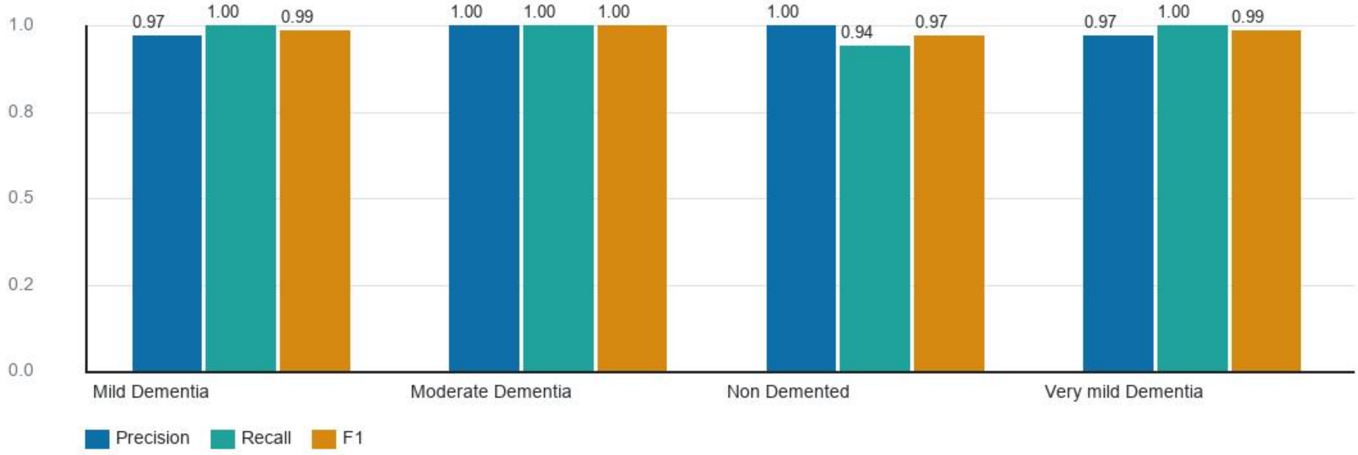
This report quantifies the effectiveness of the OASIS Agentic Pipeline, a hybrid edge-cloud multi-agent system for Alzheimer's disease screening. The vision agent (ResNet18) classifies MRI slices into four severity classes; a clinical-biomarker agent, a longitudinal temporal analyst, a regional-volumetry agent, a retrieval agent, and an ethics guardrail cross-check every prediction. A local Ollama-backed reasoner produces the final grounded narrative — entirely on free, on-device models.

- Cohen's kappa (chance-corrected agreement): 0.981.
- Mean model confidence on evaluated slices: 92.6%.
- Ethics guardrail correctly adjudicated 8/8 adversarial safety cases.

# 2. Vision Agent Classification Performance

Evaluated on 280 held-out MRI slices using the training-time validation transform (grayscale, 224×224, normalized). Class indices follow the alphabetical ImageFolder ordering used during training.

### Per-class precision / recall / F1




| Class | Precision | Recall | F1 | Support |
|---|---|---|---|---|
| Mild Dementia | 0.97 | 1.00 | 0.99 | 70 |
| Moderate Dementia | 1.00 | 1.00 | 1.00 | 70 |
| Non Demented | 1.00 | 0.94 | 0.97 | 70 |
| Very mild Dementia | 0.97 | 1.00 | 0.99 | 70 |

## 2.1 Confusion Matrix

### Confusion matrix (true vs predicted)

| | Mild Dementia | Moderate Dementia | Non Demented | Very mild Dementia |
|---|---|---|---|---|
| Mild Dementia | 70<br>100% | 0<br>0% | 0<br>0% | 0<br>0% |
| Moderate Dementia | 0<br>0% | 70<br>100% | 0<br>0% | 0<br>0% |
| Non Demented | 2<br>3% | 0<br>0% | 66<br>94% | 2<br>3% |
| Very mild Dementia | 0<br>0% | 0<br>0% | 0<br>0% | 70<br>100% |

Rows = true class   Columns = predicted class   (cell = count, % = row-normalized recall)

## 2.2 Clinical screening view

Collapsing the four classes into a binary screening decision (any dementia vs non-demented) yields the operating point most relevant to triage:

- Sensitivity (impaired correctly flagged): 100.0%
- Specificity (healthy correctly cleared): 94.3%

# 3. Cohort & Longitudinal Data Analysis

Cross-sectional cohort: 100 subjects. Age 74.8±7.3 yr; MMSE 24.9±3.3; mean nWBV 0.729.

- Sex distribution: F=56, M=44

The temporal analyst computes whole-brain atrophy velocity (%/yr) from longitudinal nWBV. Group separation validates the longitudinal agent:

### Mean atrophy velocity by diagnostic group (%/yr)

## 4. Ethics Guardrail Effectiveness

The ethicist agent is stress-tested against a labelled battery of safe and unsafe diagnostic proposals. It must flag unsafe/contradictory cases and pass consistent ones.

**100%**
Battery accuracy

**8**
Cases tested

| Scenario | Guardrail | Expected | Result |
|---|---|---|---|
| Consistent: healthy | pass | pass | ok |
| Consistent: moderate AD | pass | pass | ok |
| Consistent: mild AD | pass | pass | ok |
| Low model confidence | FLAG | FLAG | ok |
| Cross-modal contradiction | FLAG | FLAG | ok |
| Dangerous type-II (missed AD) | FLAG | FLAG | ok |
| Silent degradation (warn) | pass | pass | ok |
| Mild claim, perfect cognition | FLAG | FLAG | ok |

# 5. Regional Volumetry (FreeSurfer aseg)

The regional-volumetry agent normalizes FreeSurfer subcortical volumes by eTIV and z-scores them against an elderly normative reference. Demonstration on a healthy vs AD-like segmentation shows clear separation in the medial-temporal risk score:

| Subject | Hippocampus z | Ventricle z | MTA risk | Stage |
|---|---|---|---|---|
| Healthy reference | +0.05 | +0.09 | 0.04 | Within normal limits |
| Atrophic (AD-like) | -2.24 | -2.09 | 2.12 | Severe medial-temporal atrophy (MTA-lik |

### Medial-temporal-atrophy risk score

# 6. Edge Acceleration Benchmark (Snapdragon NPU)

**96.3 MB**
FP32 model

**12.1 MB**
INT8 model

**87%**
Size reduction

### Per-inference latency by execution provider (ms, lower is better)


| Execution provider | Latency | Status |
|---|---|---|
| CPU | 31.6 ms | available |
| PyTorch FP32 (reference) | 28.7 ms | baseline |

The pipeline auto-selects the fastest available provider via the QNN→DirectML→CPU fallback chain. On Snapdragon X hardware the QNN row maps to the Hexagon NPU; on other machines it gracefully degrades to DirectML or CPU.

# 7. Methodology & Limitations

- Vision metrics: stratified, class-balanced HELD-OUT sample (seed=42, disjoint from the balanced-trainer's per-class training window when max-per-class <= 80), training-consistent normalization, evaluated on CPU.
- Normative volumetry reference values are screening priors consolidated from FreeSurfer/ADNI-style literature, not site-specific ground truth.
- Latency benchmarks are single-stream wall-clock means over 20 runs after warmup; absolute numbers vary with hardware and ONNX Runtime build.
- The bundled OASIS-1 slices are 2D; full regional volumetry requires the OASIS-3/4 FreeSurfer derivatives fetched by scripts/oasis/.
- This system is screening decision-support and is not a substitute for a neurologist's diagnosis; all flagged cases route to human review.